

4.1 Differential Fundamentals

Differentiation is one of the most important concepts in mathematics. It allows us to determine how quickly one quantity changes with respect to another. In engineering, differentiation is used to calculate quantities such as velocity, acceleration, force, power, and flow rates.

Definition

The **derivative** of a function measures its rate of change. If

$$y = f(x)$$

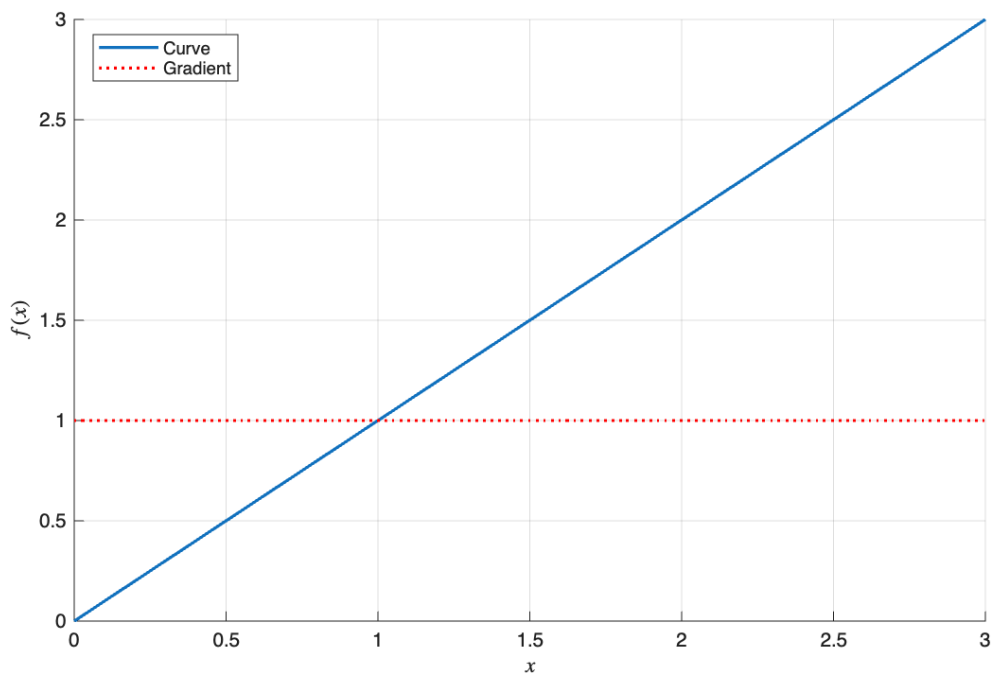
then the derivative of y with respect to x represents how rapidly y changes as x changes.

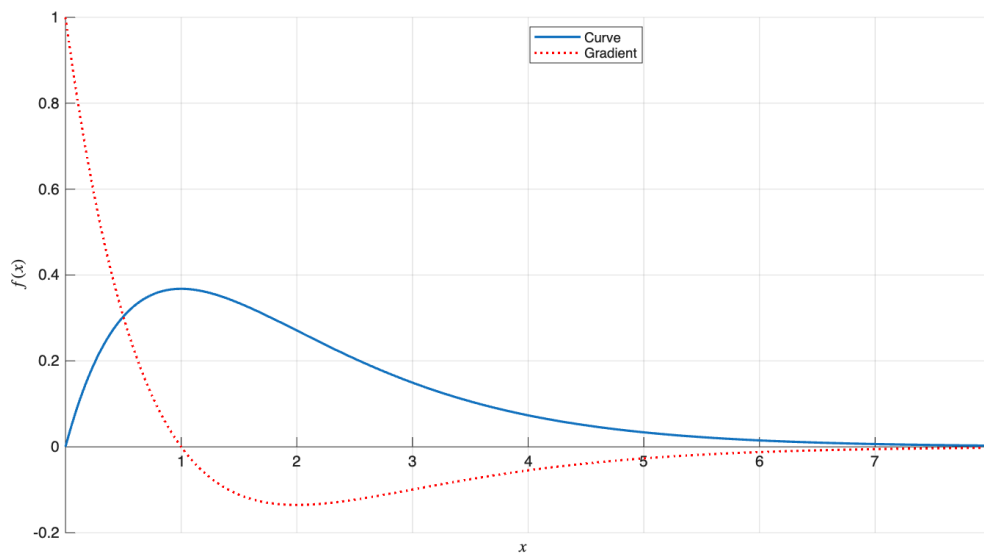
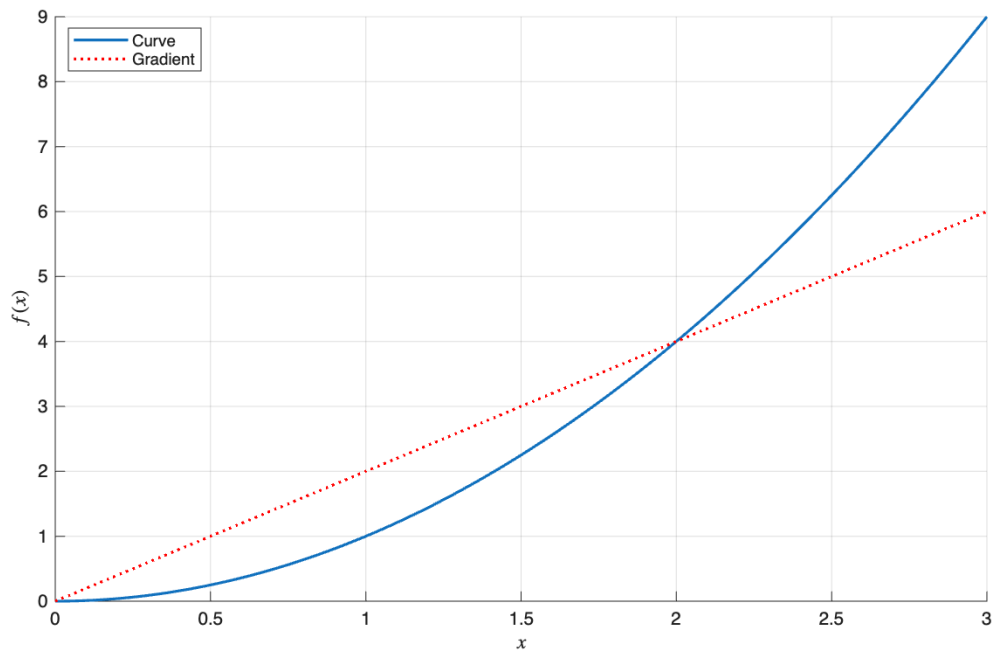
4.1.1 Gradients and Rates of Change

Recall that the gradient of a straight line is given by

$$\frac{\text{change in } y}{\text{change in } x}$$

For a curve, the gradient is not constant and generally changes from point to point.



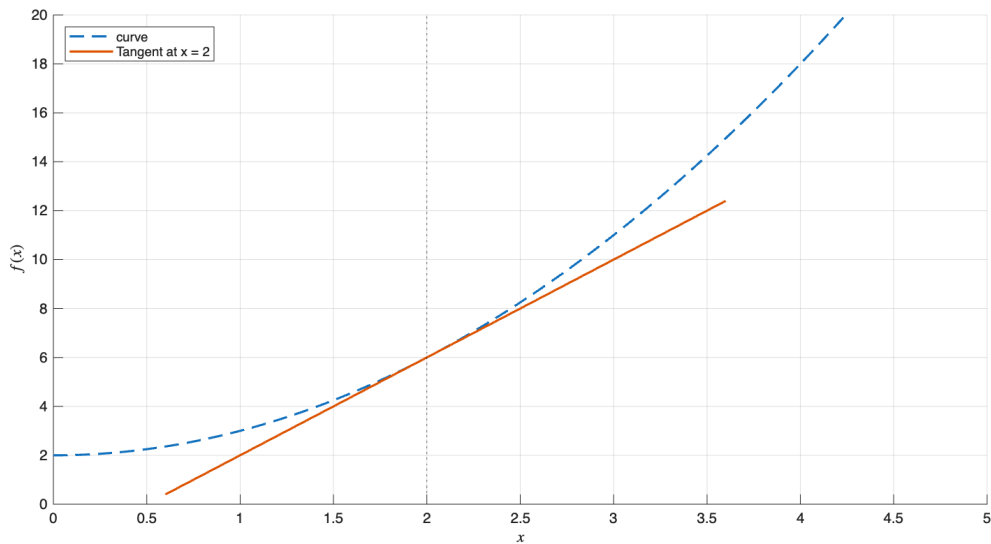


Calculating the gradient of a curve is not straightforward, as the gradient varies along the curve, i.e. it depends on x , as we can see in the images above.

Definition

The gradient of a curve at a particular point is equal to the gradient of the tangent to the curve at that point.

A tangent line is a straight line that only just touches the curve at exactly that particular point. So we could draw such a line at the point we were interested in, as shown in the figure below:



Then we would set up a triangle (as in linear cases) to calculate the gradient of the tangent. But this method is not practical.

Differentiation provides a mathematical method for determining this gradient.

4.1.2 Notation

Several equivalent notations are used for derivatives.

Definition

If $y = f(x)$, then the derivative may be written as

$$\frac{dy}{dx}$$

$$y'$$

or

$$f'(x)$$

These all represent the same quantity.

4.1.3 Standard Differentiation Rules

The following are standard rules.

Definition

For constants k , a , b and n :

Function y	Derivative $\frac{dy}{dx}$
k (any constant)	0
ax	a
ax^n	$n \times ax^{n-1}$
$a e^{bx}$	$b \times a e^{bx}$
$a \ln(bx)$	$\frac{a}{x}$
$a \sin(bx)$	$b \times a \cos(bx)$
$a \cos(bx)$	$-b \times a \sin(bx)$

The notation $\frac{d}{dx}$ is called the **differential operator**. It means “differentiate with respect to x ”.

Worked Example

If

$$y = 2x^3,$$

then

$$\frac{d}{dx}(y) = \frac{d}{dx}(2x^3)$$

This is in the form of ax^n , where $a = 2$ and $n = 3$. Therefore,

$$\begin{aligned}\frac{dy}{dx} &= 3 \times 2x^{3-1} \\ &= 6x^2\end{aligned}$$

Note that this is an expression for the gradient, which is dependent upon x .

Worked Example

Find an expression for the gradient of the curve $y = 5 \sin(2x)$.

This is in the form of $a \sin(bx)$, where $a = 5$ and $b = 2$. Therefore,

$$\begin{aligned}\frac{d}{dx}(y) &= \frac{d}{dx}(5 \sin(2x)) \\ \frac{dy}{dx} &= 2 \times 5 \cos(2x) \\ &= 10 \cos(2x)\end{aligned}$$

4.1.4 Linearity of Differentiation

Differentiation is a linear operation.

Definition

Constants may be taken outside the differential operator.

$$\frac{d}{dx}(af(x)) = a \frac{d}{dx}(f(x))$$

Note that the constants and coefficients are already taken into account within the formulae table.

Worked Example

Differentiate $y = 5x^2$.

Using the power rule,

$$\begin{aligned}\frac{dy}{dx} &= 5 \frac{d}{dx}(x^2) \\ &= 5(2x) \\ &= 10x\end{aligned}$$

Definition

Differentiation may be applied term-by-term. If

$$y = f(x) + g(x)$$

then

$$\frac{dy}{dx} = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x))$$

Worked Example

Differentiate $y = 7x^4 - 8x + 2$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(7x^4) - \frac{d}{dx}(8x) + \frac{d}{dx}(2) \\ &= 28x^3 - 8 + 0 \\ &= 28x^3 - 8\end{aligned}$$

4.1.5 Finding Gradients at Particular Points

Once the derivative has been found, gradients can be calculated by substitution.

Worked Example

Determine the gradient of $y = 4x^2$ at the point where $x = 5$.

First differentiate:

$$\frac{dy}{dx} = 8x$$

Substituting $x = 5$:

$$\begin{aligned}\left. \frac{dy}{dx} \right|_{x=5} &= 8(5) \\ &= 40\end{aligned}$$

Engineering Application

If s denotes displacement and t denotes time, then

$$\frac{ds}{dt}$$

represents the rate of change of displacement with respect to time. This quantity is called the **velocity**.

Engineering Application

If v denotes velocity and t denotes time, then

$$\frac{dv}{dt}$$

represents the rate of change of velocity with respect to time. This quantity is called the **acceleration**.

4.1.6 Exercises

Exercise

1. Differentiate:

(a) $y = 7$

(b) $y = x^5$

(c) $y = 3x^4$

(d) $y = 8x^3 - 2x + 4$

2. Determine the gradient function of:

(a) $y = 7x^{-2} + 8 \ln(5x)$

(b) $y = 5x^4 - x^2 + 3e^{6x}$

(c) $y = 6 \cos(9x) + \frac{2}{x^4} - 7$

3. Determine the gradient at the specified value of x :

(a) $y = 3x^2, \quad x = 2$

(b) $y = x^3 - 2x, \quad x = 1$

(c) $y = 4x^2 + 3x, \quad x = 5$

4. A particle moves according to

$$s = 5t^2 + 2t$$

where s is displacement (m) and t is time (s).

(a) Determine the velocity function.

(b) Determine the velocity when $t = 4$.

4.1 Solutions to Exercise 4.1.6

1. Differentiate:

(a) $y = 7$. The derivative of a constant is zero.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(7) \\ &= 0\end{aligned}$$

(b) $y = x^5$. Using the power rule:

$$\begin{aligned}\frac{dy}{dx} &= 5 \times 1x^{5-1} \\ &= 5x^4\end{aligned}$$

(c) $y = 3x^4$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(3x^4) \\ &= 4 \times 3x^{4-1} \\ &= 12x^3\end{aligned}$$

(d) $y = 8x^3 - 2x + 4$. Differentiate each term separately:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(8x^3) - \frac{d}{dx}(2x) + \frac{d}{dx}(4) \\ &= 3 \times 8x^{3-1} - 2 + 0 \\ &= 24x^2 - 2\end{aligned}$$

2. Determine the gradient function of:

(a) $y = 7x^{-2} + 8 \ln(5x)$. Differentiating term-by-term:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(7x^{-2}) + \frac{d}{dx}(8 \ln(5x)) \\ &= -2 \times 7x^{-2-1} + \frac{8}{x} \\ &= -14x^{-3} + \frac{8}{x}\end{aligned}$$

(b) $y = 5x^4 - x^2 + 3e^{6x}$. Differentiating term-by-term:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(5x^4) - \frac{d}{dx}(x^2) + \frac{d}{dx}(3e^{6x}) \\ &= 4 \times 5x^{4-1} - 2 \times 1x^{2-1} + 6 \times 3e^{6x} = 20x^3 - 2x + 18e^{6x}\end{aligned}$$

(c) $y = 6\cos(9x) + \frac{2}{x^4} - 7$. Writing the second term as $2x^{-4}$ (using the rules of indices):

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(6\cos(9x)) + \frac{d}{dx}(2x^{-4}) - \frac{d}{dx}(7) \\ &= -9 \times 6\sin(9x) + -4 \times 2x^{-4-1} - 0 \\ &= -54\sin(9x) - 8x^{-5}\end{aligned}$$

3. Determine the gradient at the specified value of x .

(a) $y = 3x^2$, $x = 2$. First differentiate:

$$\frac{dy}{dx} = 6x$$

Substituting $x = 2$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=2} &= 6(2) \\ &= 12\end{aligned}$$

Therefore, the gradient at $x = 2$ is 12.

(b) $y = x^3 - 2x$, $x = 1$. First differentiate:

$$\frac{dy}{dx} = 3x^2 - 2$$

Substituting $x = 1$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=1} &= 3(1)^2 - 2 \\ &= 1\end{aligned}$$

(c) $y = 4x^2 + 3x$, $x = 5$. First differentiate:

$$\frac{dy}{dx} = 8x + 3$$

Substituting $x = 5$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=5} &= 8(5) + 3 \\ &= 43\end{aligned}$$

4. A particle moves according to $s = 5t^2 + 2t$ where s is displacement (m) and t is time (s).
- (a) Determine the velocity function. Velocity is the derivative of displacement with respect to time.

$$\begin{aligned}v &= \frac{ds}{dt} \\ &= \frac{d}{dt}(5t^2 + 2t) \\ &= 10t + 2\end{aligned}$$

Therefore, $v = 10t + 2$.

- (b) Determine the velocity when $t = 4$. Substituting $t = 4$ into the velocity function:

$$\begin{aligned}v &= 10(4) + 2 \\ &= 42\end{aligned}$$

Therefore, the velocity at $t = 4$ is 42 m/s.